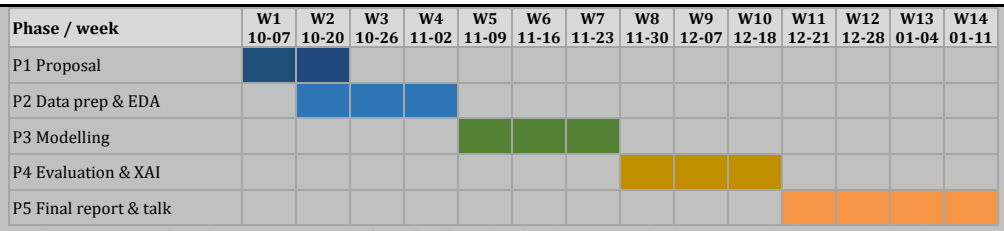
MDM2 – Case Study: Intelligent Systems in Production
One-Page Proposal

Team	Group 6
Members	Fuad Obiedat, Abdennour Djeha, Mohamed Fares, Hady Nasser
Project Title	**Early Warning of Air-Leak Failures in Metro Compressors: Explainable Anomaly Detection on the MetroPT-3 Dataset**
GitHub Repository URL	https://github.com/mohamedfarez/IntelligentSystemsCaseStudy
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Industrial Application	Railway rolling stock: predictive maintenance of the compressor Air Production Unit (APU) of a metro train (Metro do Porto).
Keywords (3–6)	predictive maintenance; anomaly detection; multivariate time series; railway; explainable AI (SHAP); air production unit
Submission Date (YYYY-MM-DD)	2027-01-12
Gant Chart	 *Weeks start on Mondays (W1 = 2026-10-07). Phase deadlines (Fridays): P1 10-16 (W1), P2 11-03 (W4), P3 11-24 (W7), P4 12-18 (W10), P5 01-15 (W14).*

1) Problem Statement & Measurable Outcomes	The Air Production Unit of a metro train feeds pneumatic subsystems such as the secondary suspension, and an unplanned failure such as,recurring air leaks takes the vehicle out of service. MetroPT-3 provides ~1.5 M multivariate sensor readings (15 signals, Feb–Aug 2020) but is unlabelled, with only four maintenance-reported air-leak failures. We will build an unsupervised or supervised, explainable early-warning model that flags abnormal APU behavior.
2) Motivation & Industrial Relevance (2–3 sentences)	Help reduce unplanned stoppages, move from fixed-interval to condition-based maintenance, for passengers higher service availability. Trains in this setting have no backup APU, so failures directly affect operations, and public benchmark data plus a demand for explainable models that maintenance staff can trust make this timely. This is relatively important for improving predictive maintenance in the transportation industry.
3) Related Work Snapshot (2–3 key references)	**[1]** Davari et al. (2021), "Predictive maintenance based on anomaly detection using deep learning for air production unit in the railway industry," IEEE DSAA. **[2]** Veloso et al. (2022), "The MetroPT dataset for predictive maintenance," Scientific Data 9:764. **[3]** Jakobs, Veloso & Gama (2025), "Interpretable Rules for Online Failure Prediction: A Case Study on the Metro do Porto dataset," arXiv:2502.07394. **Gap:** prior work centres on deep-learning detectors [1] or interpretable rules on the related MetroPT2 data [3]; we add a leakage-free, chronological benchmark of simple vs. deep detectors on MetroPT-3, scored with operational event-level KPIs (recall, lead time, false alarms) plus SHAP explanations.
4) Method & Feasibility (≤6 sentences)	**Data:** MetroPT-3 (UCI #791, CC BY 4.0; 1,516,948 rows, 7 analogue + 8 digital sensors), with labels derived from the four failure reports. **Screening/preprocessing:** check gaps and duplicates, aggregate to fixed time windows, engineer duty-cycle features (motor current, COMP, TP2/TP3), normalise, and use a chronological split (first month for training on normal data, remainder for testing, as recommended by the dataset authors). **Analysis:** compare a threshold baseline, Isolation Forest and an LSTM autoencoder (reconstruction error) and explain alarms with SHAP. **Artifacts:** reproducible notebooks in GitHub, evaluation tables/plots, final report and presentation. **Feasibility:** the 208 MB CSV is openly available and fits on a laptop, so no access or compute barriers are expected.
5) Milestones & Timeline (short table/list)	**P1 (2026-10-16, W1):** team, GitHub repository, evaluation plan and one-page proposal. **P2 (2026-11-06, W2–W4):** literature review, data download, quality checks, EDA, failure-window labelling. **P3 (2026-11-27, W5–W7):** feature engineering, baselines, LSTM autoencoder trained and tuned. **P4 (2026-12-18, W8–W10):** event-level evaluation against KPIs, SHAP explanations, preliminary results and visualisations. **P5 (2027-01-15, W11–W14):** final report, presentation and clean GitHub repository (holiday break Dec 21–Jan 3).
6) Risks & Ethics (1–2 sentences)	Only four labelled failures (all air leaks) limit statistical power and generalisation, so we will use chronological splits, event-level metrics and report uncertainty rather than over-claim. The data are machine sensor readings without personal information; the dataset is cited per its CC BY 4.0 licence, and all references and code reuse will be attributed.